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21 March 1979
SUPERSEDING
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MILITARY SPECIFICATION

CAMERA, COPYING, LITHOGRAPHIC, 40 BY 40 INCHES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a process copying camera used for precision copying of photolithographic material.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-F-336	- Film, Photographic, Photomechanical Black-and-white.
CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).
TT-L-20	- Lacquer, Camouflage.

FSC 3610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant, (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514	- Plates Identification, Instruction and Marking, Blank.
MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3615	- Cameras, Copying, Lithographic, Packaging of.
MIL-D-4827	- Developer, Photographic, Paper, Armed Forces Developer No. 25.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-P-15024/6	- Plate, Identification.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-150	- Photographic Lenses.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

OCCUPATIONAL SAFETY AND HEALTH ACT OF 1970 (OSHA)

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The camera shall consist of a rigid overhead truss frame and supports, a camera back, a lensboard carriage, and a copyboard carriage with an interchangeable transparency holder, vacuum copyboard, and a wooden copyboard. The lensboard and copyboard carriages shall be supported on rails which are mounted on the overhead truss. The optical

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axis of the camera shall be horizontal and shall be 52 inches (1321 millimeters (mm)), plus or minus 2 inches (51 mm), above the floor.

3.2 First article (first produced camera). The contractor shall furnish a camera for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce cameras that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.7).

3.4 Safety. All rotating or reciprocating parts or parts subject to high temperature shall be guarded in accordance with current OSHA regulation, 29CFR, Chapter XVII, 1910.212 when such parts are exposed to contact by personnel or otherwise create a hazard.

3.5 Human Factors. The design of the camera shall allow for operation by Army personnel both men and women from 5th to 95th percentile as defined in MIL-STD-1472.

3.5.1 Noise limits. The noise of the vacuum pump shall not exceed the limits specified in MIL-STD-1474, Category D for steady-state noise when tested in accordance with 4.5.2.1.1.

3.6 Capacities and dimensions. The camera shall have a capacity to handle negatives from 4 by 5 inches (102 by 127 mm) to not less than 40 by 40 inches (1016 by 1016 mm). The vacuum copyboard shall be not less than 48 by 60 inches (1219 by 1524 mm), the wooden copyboard shall be not less than 48 by 72 inches (1219 by 1829 mm), and the transparency holder shall be not less than 40 by 40 inches (1016 by 1016 mm). The overall dimensions of the camera shall be not more than 26 feet (7.9 meters (m)) in length, 14 feet (4.3 m) in width, and 10 feet (3 m) in height.

3.7 Overhead truss and supports. The structure supporting the truss and its components shall be steel, designed to be floor mounted, and shall be braced to support all components of the camera under all conditions of use. Suspension of the truss shall be through vibration

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mounts, with mechanical snubbers to prevent excessive movement of the various components. The vibration mounts shall absorb all transmitted vibrations and normal shock impulses. The truss shall be a rigid, tubular steel structure, having sufficient strength to support the various components of the camera. Structures other than tubular may be used if they have equal or greater torsional and flexural rigidity, and otherwise provide the structural characteristics of a tubular design. The truss shall be provided with rails which are adjustably supported and secured to the truss. The rails shall rigidly support and maintain the alinement of the lensboard and copyboard carriages. The rails shall be sufficiently strong to guide the carriages without excessive wear or permanent deformation resulting from high-load concentrations. The rail shall not wear more than 0.0005 inch (0.013 mm) when the copyboard is moved back and forth on the carriages for 100 cycles.

3.8 Camera back. The camera back shall be a rigid metal structure, secured and braced to maintain accurately the negative plane at an angle of 90 degrees plus or minus 15 seconds, to the bearing surfaces of the rails. The camera back shall support the ground-glass and vacuum-back assemblies. It the movement of any component requires the lifting of weights, that component shall be counterbalanced. Locking devices shall be provided to lock each component in exposure position.

3.8.1 Ground glass. The ground glass shall be housed in a rigid metal frame that is hinged to pivot about a vertical axis at one side of the camera back. As an alternative, the ground glass in its frame may be hinged and counterbalanced above an overhead horizontal axis, or may slide out of its position of use provided the required accuracy is retained. The grained surface shall allow resolution of not less than 20 lines per millimeter. When the ground glass is locked in the exposure position, the grained surfaced shall face the lens and shall be in the focal plane of the camera within 0.003 inch (0.076 mm). The hinged or sliding arrangement of the ground glass shall permit ready access to the ground surface for attaching tape markers and for marking dimensions. The grained surface shall have a 1 inch (25.4 mm) clear square or circle in its center and 0.25 inch (6.35 mm) clear diagonals and shall have readily visible, vertical and horizontal, etched or clear glass scales that are numbered to indicate negative size in full inch dimensions (to read correctly when viewed through the glass from the back of the camera). The edges of the ground glass shall be ground square and true to standard dimensions to facilitate replacement. The thickness of the ground glass shall be 0.25 inch (6.35 mm).

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3.8.2 Vacuum back. The vacuum back shall pivot about a vertical axis to the side of the camera back not common with that of the ground glass. The film-holding surface of the vacuum back 4 by 5 inches (102 by 127 mm) to 40 by 40 inches (1016 by 1016 mm) shall be a plane, flat within plus or minus 0.003 inch (0.076 mm). When locked in exposure position, the film-holding surface shall be parallel to the copyboard surface within 0.006 inch (0.152 mm). The vacuum back shall have a black or neutral low-light-reflection color. The film-holding surface may be perforated or porous; however, any openings shall be sufficiently small that they do not distort the surface of thin base film. Vacuum zoning control shall be provided to hold, without mechanical aids, any standard size film (whether in sheet form or cut from a roll) within the range specified. Zoning control shall be by a simple valve and chamber arrangement, and shall permit rapid selection of the desired zone under darkroom safelight conditions. The zoning shall also provide for the use of standard commercial sizes of contact printing screens. Rectangles corresponding to standard film sizes and contact screens shall be marked on the holding face (long dimension horizontal) and shall be numbered for size. The marking shall be readily visible under darkroom conditions and shall not cause surface distortion of thin base film.

3.8.2.1 Vacuum pump. A vacuum pump and motor shall be furnished for the vacuum back. The pump shall be either a multi-stage turbine or an oilless, graphited-vane, rotary pump of adequate capacity to hold the specified sizes and types of film. The vacuum pump shall be furnished complete with base, drive switches, controls, fittings, vacuum gage, exhaust muffler, vibration mounts, and connecting hose.

3.9 Carriages. The lensboard and copyboard carriages shall rigidly support their components in a plane perpendicular to the optical axis of the camera throughout their range of travel. The carriages shall be electric-motor-driven, with a fast and slow movement, and shall be pushbutton controlled from both the camera room and the darkroom. The drives shall accurately position either carriage within 0.020 (0.51 mm) inch of any predetermined position within the travel limits of the camera. Provision for take-up of backlash shall limit such movement to less than 0.005 (0.127 mm) inch. Manual controls for temporary use in the event of power failure shall also be provided. Operating controls in the darkroom shall be placed in a position which permits the operator to use the controls while observing the image on the ground glass. Counters shall be provided in the darkroom and the camera room to

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indicate the distances of the carriages from the exposure plane in increments of 0.001 inch (0.025 mm). The counters shall be coupled to the carriage drive mechanisms, shall be manually and motor-driven, and shall be rated for continuous operation of at least 5,000 figures per minute. The counters located in the darkroom shall be illuminated by ruby lighting, directed and shielded so that the light will not reach the negative during exposure and normal handling. Limit switches shall be provided to prevent the carriages from overrunning the track or from bumping against each other as a result of either hand or motor-driven operation. The copyboard carriage shall also be controlled by facilities inside and outside the darkroom for rotating the copyboards through an angle of at least plus or minus 3 degrees from vertical, such rotation to be approximately centered on the optical axis of the camera. Safeguards shall be provided to prevent jamming at the rotation extremes.

3.9.1 Lensboard. A lens mount shall be provided on the lensboard to permit interchangeability of lenses (see 3.12). The mount shall rigidly support the lens in proper position without play. The lensboard shall include a motor or manual drive to displace the lens mount up to not less than 3 inches (76 mm) in any vertical or lateral direction from the camera optical axis. Controls for moving the lens mount shall be located in the darkroom in a position convenient to the operator while the operator is observing the ground glass. A solenoid-operated shutter of sufficient diameter to encompass the field of coverage of any of the lenses shall be provided. The shutter shall be mounted on the back of the lens mount inside the bellows. The shutter shall be provided with a current rectifier to prevent vibration. Shutter operation shall be by line control switches located in the darkroom and in the camera room. When specified (see 6.2), provision shall be made to operate the shutter by an interval timer or a light-integrating device. The lensboard and the lens mount shall be completely light-tight in any position of the mount.

3.9.1.1 Flash lamp. The lensboard assembly shall be provided with a spring-counterbalanced, pivoting flash lamp arranged so that the flash lamp can be quickly and positively brought in front of the lens for exposure for any position of the lens or tilted out of range of the optical system as required. The flash lamp shall incorporate a diffused light source containing a 60-watt bulb, and shall have a line control switch.

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3.9.2 Copy holders. The copyboard carriage shall be provided with three interchangeable copy holders, a transparency holder, a vacuum copyboard, and a wooden copyboard. When attached to the carriage, the photographic planes of the three copy holders shall be identical within 0.003 inch (0.076 mm) of each other. The carriage shall support any copy holder with the center of the copy holder in the approximate optical axis of the camera. All copyboard photographic planes shall be parallel to the exposure plane at any position along the camera ways and with any weight of copy up to 35 pounds (15.88 kilograms (kg)).

3.9.2.1 Transparency holder. The transparency holder shall consist of a metal frame supporting a pair of positioning bars that are adjustable so as to center and hold for photographing, glass or film transparencies from 4 by 5 inches (102 by 127 mm) to 40 by 40 inches (1016 by 1016 mm). Indicating scales and an adjustable side stop on the lower bar shall be provided to aid in centering the transparency to the optical axis. Illumination for exposure shall utilize the camera lamps placed behind an attached masking screen or hood to confine the exposure area to the maximum transparency capacity. A flashed opal diffusing glass, or 0.125-inch (3.2 mm) thick plastic sheet with equivalent color and uniformity of light transmission, shall extend beyond the maximum transparency size and shall be supported in a suitable frame. When specified (see 6.2), the masking screen or hood shall be replaced with an integral light box attached to the copyboard structure and containing a closely spaced bank of 40-watt, rapid-start, cool-white, fluorescent lamps, together with ballasts, covered with a diffuser equal to that specified above. The integral light source shall be designed to facilitate service and use, and shall provide a uniform field of illumination over the maximum transparency area.

3.9.2.2 Vacuum copyboard. The face of the vacuum copyboard shall be a plane within plus or minus 0.003 inch (0.076 mm), and shall support copy in a position perpendicular to the optical axis of the camera throughout its range of travel. The copyboard vacuum shall be zone-controlled to hold any size of copy from 8 by 10 inches (203 by 254 mm) to not less than 48 by 60 inches (1219 by 1524 mm) without masks or mechanical aids. Zoning control shall be a simple control valve. The holding surface may be perforated or porous; however, any openings shall be sufficiently small that they will not visibly indent paper copy of 20-pound (9.1 kg) bond basis (equivalent to 107-pound (48.5 kg), 25 by 40 inch (635 by 1016 mm), 1,000 sheet basis). Suction shall be sufficient to hold flat all copy sizes and materials up to 0.125-inch (3.2 mm) thick aluminum or 0.250-inch (6.35 mm) thick hardboard. The

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vacuum copyboard sides shall be free of projections (except the overhead structure) to permit mounting larger copy than 48 by 60 inches (1219 by 1524 mm). The vacuum copyboard holding surface shall be finished a lusterless black or dark neutral color resistant to scratching or peeling. Vertical and horizontal centerlines shall be scribed across the face of the holding surface.

3.9.2.2.1 Vacuum pump. A vacuum pump and motor shall be provided for the vacuum copyboard. The pump shall be either a multi-stage turbine or an oilless, graphited-vane rotary pump of adequate capacity to hold the specified size and type of copy. The vacuum pump shall be furnished complete with base, drive, switches, controls, fittings, vacuum gage, exhaust muffler, vibration mounts, and sufficient connecting hose to follow the vacuum copyboard throughout its range of travel.

3.9.2.3 Wooden copyboard. The wooden copyboard shall be 0.625-inch (15.88 mm) thick and shall be fabricated of plywood or glued strips of thoroughly-seasoned pine. The overall size shall be not less than 48 by 72 inches (1219 by 1829 mm). The wooden copyboard shall be mounted on a lightweight metal structure to retain a flatness varying not over 0.031 inch (0.79 mm) and to permit expansion and contraction of the wood. The wood shall be treated with a sealing compound and shall be painted a lusterless black. Vertical and horizontal centerlines shall be scribed on the face.

3.10 Accuracy. All operating planes shall be within 0.005 inch (0.127 mm) of parallelism and alinement for the entire range of movement on the camera. An additional tolerance of 0.005 inch (0.127 mm) is permitted on the wooden copyboard.

3.11 Rigidity. The camera shall be sufficiently rigid to meet the following requirements.

3.11.1 Vibration. With an image projected onto the ground glass surface and with both vacuum pumps in operation, the vibration of the image margins shall not exceed 0.0005 inch (0.013 mm) in any direction.

3.11.2 Displacement of components. The shift of an image projected onto the ground-glass surface shall not exceed 0.002 inch (0.051 mm) from movement of vacuum back. The image shift caused by truss deflection when the vacuum copyboard is placed on the copyboard carriage shall not exceed 0.004 inch (0.102 mm).

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3.11.3 Shock. When a weight of approximately 20 pounds (9.1 kg) is secured to a rope tied to the side member of the copyboard structure and dropped 2 feet (610 mm) to the end of the length of rope, the vibration of the image margins of an image projected onto the ground glass surface shall be dampened to 0.0005 inch (0.013 mm) within 10 seconds and the image shall return to within 0.0005 inch (0.013 mm) of its original position.

3.12 Lenses. Two interchangeable, apochromatic, process lenses conforming to MIL-STD-150, Type IV, shall be provided with each camera. The nominal focal lengths for the lenses shall be 24 inches (610 mm) and 42 inches (1067 mm). Each lens shall be mounted in a barrel which is mounted on a lens plate to permit interchangeability of the lenses. Each lens shall be provided with a lens cap, and each lens with its mounting plate shall be housed in a hinged and lined box with slots or compartments for the filter holders. A list shall be fastened to the interior of the lens box showing the following information for each lens: Lens serial number, equivalent focal length, front focal length, back focal length, and nodal point separation.

3.12.1 Range of reproduction. The 42-inch (1067 mm) lens shall provide a range of reproduction of from at least 1/3 to 3 times the copy size. The other lens shall have the greatest range of reproduction possible under the range specified for the 42-inch (1067 mm) lens.

3.12.2 Covering power. The half angle of coverage for each lens shall be not less than 22.5 degrees.

3.12.3 Distortion. The amount of distortion for any lens shall be not more than 0.3 mm at any point in the field.

3.12.4 Resolving power. The resolving power in terms of lines per millimeter in the image plane shall be not less than that shown in Table I. The resolving power is defined as the greatest number of lines per millimeter resolved in that orientation of lines (radial or tangential) having the poorest definition.

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TABLE I. Resolving Power.

Lens			
42-inch (1067 mm)		24-inch (610 mm)	
Angle from center	Resolving power	Angle from center	Resolving power
(degrees)		(degrees)	
0	23	0	23
5	20	5	20
11	20	11	20
16	12	16	12
22	12	22	12

3.12.5 Color correction. Color correction shall be sufficient to produce equally sharply-defined negatives over the range of the visible spectrum, and to make visual and photographic focus identical or within 0.01 inch (0.254 mm) of each other. A point image projected onto the ground glass with white light shall be displaced no more than plus or minus 0.1 mm when colored light within the visible spectrum is used.

3.12.6 Filter holders. Four, split-stop, gelatin filter holders shall be provided for each lens. The filter holders shall be designed to be inserted into slots in the lens barrel.

3.12.7 Aperture stops. Each lens shall be equipped with a continuous diaphragm control, and shall include, but shall not be limited to, the following stops: f/11, f/16, f/22, f/32, f/45, and f/64.

3.12.8 Focusing charts. The camera shall be calibrated for each 1 percent enlargement or reduction for each lens throughout the range of the camera with the lens in proper focus at all settings. Three charts shall be provided for each lens showing counter readings of the lensboard and copyboard carriages to 0.001 inch (0.025 mm) for each 1 percent enlargement and reduction for the full range of each lens. The charts

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shall have the serial number of the lens, focal length, nodal separation, camera serial number, and other characteristics printed on the face. Each chart shall be provided with a protective holder.

3.12.9 Lens coating. The lenses shall be coated with a reflection reduction coating in accordance with MIL-C-675.

3.13 Bellows. The camera shall be equipped with a ruggedly-constructed bellows with removable sections to provide for the full range of lens movement, without image obstruction, in any position of the lens. The corners of the bellows shall be reinforced with durable leather or plastic material. The entire bellows including the joints between the sections shall be light-tight. The bellows shall not reveal light visible in the darkroom. The bellows shall not sag when extended or twist when contracted.

3.14 Pulsed xenon lamps. Two sets of double-deck, motor-driven, high-intensity pulsed xenon lamps shall be provided to illuminate the copyboards. The lights shall be mounted on two counterbalanced floor stands, with two lights on each floor stand, mounted one above the other. The floor stand shall occupy a minimum of floor space and shall be mounted on not less than three rubber-tired casters. The stand shall have an overall height of not less than 8 feet (2438 mm). A locking mechanism, preferably foot operated, shall be incorporated in the base or as part of the casters to prevent the stand from rolling on irregular or sloping floors. The lights shall be mounted with all movements completely independent of each other. Positive locking devices shall be provided for all movements. Vertical movement shall be provided to place the center of the upper lamp from 68 inches (1727 mm) to 93 inches (2362 mm) above the floor and the center of the lower lamp from 23 inches (584 mm) to 36 inches (914 mm) above the floor. A tilting adjustment shall permit tilting either lamp at least 15 degrees above and 35 degrees below the horizontal position. The lights shall be connected to be operated from a single power regulating unit. The intensity of the lamps shall be not less than 940 foot-candles and not less than 185 foot-candles using an ultraviolet filter; the average intensity of light at any point on the plane testing surface shall not deviate more than plus or minus 15 percent from the average illumination of the plane surface, and the continuity of intensity of light at any point shall not vary by more than plus or minus 5 percent from the average light intensity at that point during the life of a single lamp.

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3.15 Darkroom wall section. The camera shall be equipped with a metal or wooden wall section to provide a light-tight fit between the camera and the wall of the darkroom. An external rectangular frame, and material and supplies for joining the frame to the darkroom wall to produce a light-tight flexible connection, shall also be provided.

3.16 Electrical requirements. The electrical components of the camera shall be designed for operation on 120/208-volt, 3-phase, 4-wire, 60-Hertz current. Fractional horsepower motors and controls may be operated as single-phase equipment. Motors shall conform to CC-M-1807, continuous duty, as applicable. All motors shall be equipped with thermal overload switches, and all wiring shall conform to NFPA No. 70.

3.17 Electromagnetic interference. The electromagnetic interference emission characteristics of the camera shall conform to MIL-STD-461, Class 11B.

3.18 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.18.1 Lubricants. The camera shall operate as specified herein when lubricated with military lubricants conforming to MIL-G-23827 and MIL-G-10924.

3.18.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, Type I, II, or III. Fittings shall be located in a protected position. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

3.18.3 Delivery lubrication service. When delivered, all components requiring lubrication, except those for which preservatives are specified in MIL-C-3615, shall be lubricated. Grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-23827 or MIL-G-10924, as applicable. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of greases used.

3.19 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition

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I or II, as applicable (see 3.19.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.19.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.20 Identification marking. The camera shall be identified in accordance with MIL-STD-130. The marking shall be applied to the camera on plates conforming to MIL-P-15024/6. Each plate shall be attached in a location where the plate will be both visible and legible.

3.21 Instruction plates. Each camera shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the camera. Instruction plates shall conform to MIL-P-514, Type III, Composition C, or Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.22 Treatment and painting. All portions of the camera normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, except that copyboards, lensboard, camera back, bellows frames, darkroom connecting frame, and all other parts of the camera that may reflect light into the lens or onto the negative shall be coated with lacquer conforming to TT-L-20, Color Black, Number 37038.

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3.23 Toolbox. The equipment shall be provided with a toolbox constructed of metal which has a thickness of not less than 0.049 inches (1.25 mm) and shall be of sufficient size to accommodate maintenance tools, operating tools and spare parts.

3.24 Workmanship. All parts, components, and assemblies of the camera including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.24.1 Steel fabrication. Steel used in the fabrication of the camera shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having an eroded surface is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Burned surfaces of flame-cut material shall be free of slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat treatment is required. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced camera inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced camera inspection.

4.3.1 Examination. The first produced camera shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced camera shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each camera shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each camera shall be tested as specified in 4.5.2.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The camera shall be examined as specified herein for the following defects:

101. Safety devices missing or not as specified.
102. Capacities and dimensions not as specified.
103. Components missing or not as specified.
104. Identification or special marking missing, incomplete, or illegible.
105. Treatment and painting not as specified.
106. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Human Factors. The human factor requirements shall be evaluated during testing. Nonconformance to 3.5 shall constitute failure of this test.

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4.5.2.1.1 Noise level measurement.

4.5.2.1.1.1 Representative duty station. Measure the noise level at every 45 degrees plus or minus 5 degrees of azimuth, 65 inches plus or minus 1 inch above ground level, at a 5-foot radius from the geometric center of the vacuum pumps in accordance with MIL-STD-1474 for steady-state noise. The vacuum pumps shall be operated at rated capacity when the copy and film holding boards are set to the maximum holding area. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.2 Electromagnetic interference. The first produced camera shall be tested to determine conformance to 3.17. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.3 Structural rigidity.

4.5.2.3.1 Vibration. A reticuled microscope having not less than 20 power magnification, with 0.001 inch (0.025 mm) graduations, shall be attached to the back of the ground glass and focused on the grained surface. Using the 24-inch (610 mm) lens positioned for 1/2 size reproduction and a half-tone negative or print on the copyholder (or other suitable copy having detail measuring 0.001 (0.025 mm) or 0.002 inch (0.051 mm) in line-width), bring the image of the detail to coincide with one of the divisions on the reticule. Both vacuum pumps shall be operated. Nonconformance to 3.11.1 shall constitute failure of this test.

4.5.2.3.2 Displacement of components. While viewing the detail as in 4.5.2.3.1, swing the vacuum back from the extreme nonuse position to a location immediately behind the observer. Repeat this test procedure by suspending a weight approximately equal to the weight of the vacuum copyboard to the copyboard frame structure. Nonconformance to 3.11.2 shall constitute failure of this test.

4.5.2.3.3 Shock. While the image is observed in the reticuled microscope as in 4.5.2.3.1, the weight, specified in 3.11.3, secured to a rope tied to the side member of the copyboard structure, shall be dropped the distance specified in 3.11.3 to the end of the length of rope. Nonconformance to 3.11.3 shall constitute failure of this test.

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4.5.2.4 Accuracy.

4.5.2.4.1 Parallel photographic planes. The image of a grid on the copyboard shall be superimposed over an identical grid on a ground glass and observed for reproduction sizes including at least 2x, 1x, 0.5x, 0.4x, 0.3x, and 0.2x, using the 24-inch (610 mm) lens. Both grids shall have a gridded area of not less than 20 by 20 inches (508 by 508 mm), and shall be sectioned with lines not exceeding 0.003 inch (0.076 mm) wide and spaced at intervals of 1.000 inch (25.4 mm) or decimal fraction thereof. The copyboard grid may be photographically printed either on a ground glass for use as a transparency, or on a white lacquered metal plate for use on the vacuum copyboard. The observing grid shall be photographically printed on a grained plate glass capable of resolving at least 20 lines per millimeter. Image agreement of the maximum grid squares for the above reproduction ratios with the grid on the ground glass shall permit slight displacement that does not become a visual separation of the superimposed lines when viewed under a 5 power or greater magnifier. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.4.2 Interchangeability of copy planes. Tests similar to that made in 4.5.2.4.1 shall be made to determine the interchangeability of the copyboards. Nonconformance to 3.9.2 shall constitute failure of this test.

4.5.2.4.3 Flatness of planes. Flatness of the vacuum back, the ground glass, and the copyboards shall be determined with these components in operating position on the camera. A heavy precise straight-edge in combination with feeler gages, or a traveling micrometer indicator shall be used to test the plane along both diagonals, the horizontal and vertical centerlines, and the margins within two inches (50.8 mm) of their rated size capacities. Nonconformance of the vacuum back, the ground glass, the vacuum copyboard, and the wooden copyboard to 3.8.1, 3.8.2, 3.9.2.2, or 3.9.2.3 shall constitute failure of this test.

4.5.2.4.4 Carriage location accuracy. The accuracy of carriage position as indicated by the counters shall be tested at distances of 6 (1.8 m), 9 (2.7 m), 12 (3.7 m), and 15 feet (4.6 m) between the image plane on the ground glass and the vacuum copyboard surface by using any precision gage, or distance pieces, having a cumulative error not more than 0.005 inch (0.127 mm). The camera shall be at a temperature of 70° F (21° C) plus or minus 3° F (1.6° C), or if this condition is

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not obtainable at the time of inspection, compensation shall be allowed for camera expansion and expansion of gages or distance pieces. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.5 Lens tests.

4.5.2.5.1 Focusing system. The camera shall be set for reproduction with the 42-inch (1067 mm) lens. Copy not less than 40 inches (1016 mm) square consisting of fine black grid lines not more than 0.003 inch (0.076 mm) in line-width (may be composed if necessary of several sections of photographically reproduced or printed grids) shall be placed on the copyboard and its image shall be focused on the ground glass using the counters. With the lens at f/16, the image shall be examined with a magnifier of not less than 5 power magnification. While observing the grid lens, move the copyboard slightly away from the lens and also toward the lens to determine whether such movement increases the overall definition, denoting possible incorrect counter readings. The correctness of the counter readings shall also be determined for the reproduction ratios of 2x, 1x, 0.5x, 0.4x, and 1/3 size by slight displacements of the lensboard, both forward and reverse. This procedure shall be repeated with the 24-inch (610 mm) lens. Nonconformance to 3.12.8 shall constitute failure of this test.

4.5.2.5.2 Field of coverage. Using the same copy, lens aperture, and reproduction ratios used in 4.5.2.5.1, examine the image on the ground glass with a magnifier of not less than 5 power magnification for uniform definition of the grid lines. Inability to acquire a definition of the grid lines of less than 22.5 degrees for each lens shall constitute failure of this test.

4.5.2.5.3 Distortion. Test for distortion of each lens in accordance with MIL-STD-150, Method 30, with the lens at f/11 and the camera set for the same size reproduction. The light source shall be the camera's pulsed xenon lamps. Distortion more than 0.3 mm at any point in the field shall constitute failure of this test.

4.5.2.5.4 Resolving power. Test for resolving power of each lens in accordance with MIL-STD-150, Method 12. The tests shall be made with the camera set for the same size reproduction and with the lenses at f/22. The light source shall be the camera's pulsed xenon lamps. The test targets shall be a high-contrast type with dark lines on a white background. The film used shall be in accordance with L-F-336, designator 51-44-35c, shall be exposed for optimum results, and shall be

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developed for 1 minute and 15 seconds at 68° F (20° C) in developer conforming to MIL-D-4827 diluted with 5 parts water. Nonconformance to the resolving power specified in Table I shall constitute failure of this test.

4.5.2.5.5 Color correction. The following tests for color correction shall be performed for all lenses. In lieu of these tests, a certified report by the National Bureau of Standards showing that the individual lenses comply with the color correction requirements specified will be accepted.

4.5.2.5.5.1 Visual and photographic focus. A section of a grid about 1 by 6 inches (25.4 by 152 mm), with black lines on a white background, shall be placed at the center of the copyboard. The lines shall be not more than 0.003 inch (0.076 mm) in width (a larger grid may be masked with black paper to give the correct size). The camera shall be set for 1/2 size reproduction. A series of seven adjacent line exposures shall be made on a strip of film conforming to L-F-336, designator 51-44-35c, by using the lensboard movement to displace the image approximately 0.500 inch (12.7 mm) between exposures, noting the direction of progression on the film. After each exposure the lensboard carriage shall be moved forward 0.01 inch (0.254 mm), the first carriage setting being 0.3 inch (7.62 mm) less than the specified counter setting and the last setting being 0.03 inch (0.762 mm) more than the specified counter setting. The film shall be developed and processed according to the manufacturer's instructions. Nonconformance to 3.12.5 shall constitute failure of this test.

4.5.2.5.5.2 Longitudinal chromatic aberration. Using the copy specified in 4.5.2.5.5.1, a strip of film conforming to L-F-336, designator 51-44-35c shall be successively exposed through red, green, and blue gelatin filters inserted in the lens (Wratten 25, 58, and 46B, or equivalent) using the lensboard movement to displace the images and using the specified counter settings for 1/2 size reproduction. One unfiltered control exposure shall also be made, which will provide a total of four adjacent exposures on the film. Compensations in exposure time shall be made to accord with the respective filter factors. The film shall be developed and processed according to the manufacturer's instructions. The filtered images should be equally sharply defined. If poor definition was obtained by the use of any filter, that filter shall be used for a further test to determine if sharp definition can be obtained by moving the lens carriage no more than 0.01 inch (0.254 mm) either

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forward or reverse, using the carriage moving procedure given in 4.5.2.5.5.1. Nonconformance to 3.12.5 shall constitute failure of this test.

4.5.2.5.5.3 Lateral chromatic aberration. Tests to determine the lateral chromatic aberration shall be in accordance with MIL-STD-150, Method 20 or Method 21. Nonconformance to 3.12.5 shall constitute failure of this test.

4.5.2.6 Light-tight construction. Tests for light-tight construction of the lensboard components, the bellows, and the darkroom wall section shall be conducted by closing the lens shutter or capping the lens, moving the lensboard carriage position forward to extend all bellows to at least one-half their maximum extension, directing the camera pulsed xenon lamps toward the lensboard, bellows, and wall section, and then turning off all darkroom lights for at least five minutes to allow the eyes to become dark-adapted. Then the pulsed xenon lamps shall be turned on and observations shall be made all around the darkroom wall section and inside the bellows for evidence of light leakage. The lensboard shall then be moved to all lateral and vertical extremes, observations again being made for evidence of light leakage. Nonconformance to 3.8.1, 3.13, or 3.15 shall constitute failure of this test.

4.5.2.7 Durability of rails. The durability of the rails to deformation due to metal flow or wear shall be determined by running the copyboard carriage back and forth for 100 cycles over a 6 to 10 inch (152 to 254 mm) length of the rail. The rail thickness shall be measured to 0.0001 inch (0.003 mm) before and after the tests. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.8 Limit switches and safety devices. All limit switches and safety devices shall be tested to determine proper operation. Due caution shall be observed as warning plates instruct. Nonconformance to 3.9 and 3.16 shall constitute failure of this test.

4.5.2.9 Holding ability of vacuum surfaces. A piece of film 40 by 40 inches (1016 by 1016 mm) cut from a 40-inch (1016 mm) roll, a piece of film 4 by 5 inches (102 by 127 mm) cut from a 30-inch (762 mm) roll and a sheet of commercial ortho film (0.008-inch (0.203 mm) thick) shall be successively placed on the vacuum back. A 30 by 40 inch (762 by 1016 mm), 0.125-inch (3.175 mm) thick aluminum plate and a 48 by 48 inch

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(1219 by 1219 mm), 0.250-inch (6.35 mm) thick hardboard (such as plywood or masonite) mount shall be successively placed on the vacuum copyboard. Nonconformance to 3.8.2 or 3.9.2 shall constitute failure of this test.

4.5.2.10 Light intensity and uniformity tests. The light shall be positioned 52 inches (1321 mm) from a plane surface centered directly in front of and parallel to the reflector opening. The area of the plane surface shall be 28 by 36 inches (711 by 914 mm) with the long dimension horizontal. Measurements of light intensity shall be made with a foot-candle meter, which shall be equipped with a probe having a quartz filter over the light cell, and shall be sensitive to both visible and ultraviolet light. Readings shall be made with both the bare probe and with a Corning No. 7059 or equal ultraviolet filter over the probe. Readings shall be taken with the probe in the center, at each corner, and at the midpoint of each side of the plane surface. Not less than three sets of readings shall be taken for each filter. The intensity readings for the bare probe and the filtered readings shall be averaged separately. Nonconformance to 3.14 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking for Level A or Level B shall be examined to determine compliance with the quality assurance provisions of MIL-C-3615. The preservation, packing, and marking for Commercial shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.1.1 Level A. Each complete camera shall be preserved in accordance with the Level A requirements of MIL-C-3615.

5.1.2 Level B. Each complete camera shall be preserved in accordance with the Level B requirements of MIL-C-3615.

5.1.3 Commercial. Each complete camera shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

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5.2.1 Level A. Cameras, preserved as specified in 5.1, shall be packed in accordance with the Level A requirements of MIL-C-3615.

5.2.2 Level B. Cameras, preserved as specified in 5.1, shall be packed in accordance with the Level B requirements of MIL-C-3615.

5.2.3 Commercial. Cameras, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military packaging shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The camera is intended primarily for use by base topographic units for the precise reproduction of map originals and photo-maps.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the first produced camera (see 3.2).
(c) When an interval timer or light-integrating device is required (see 3.9.1).
(d) When an integral light box is required (see 3.9.2.1).
(e) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced camera. Any changes or deviations of production cameras from the approved first produced camera during production will be subject to the approval of the contracting officer. Approval of the first produced camera will not relieve the contractor of his obligation to furnish cameras conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional

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materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each camera.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of cameras.

6.6 Noise limits. When the noise level of the vacuum pumps exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the vacuum pumps, the distance from the vacuum pumps at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

6.7 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-C-20654C Camera, Copying, Lithographic, 40 by 40 Inches

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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